



Physics Worksheet

Derivative Practice

Name: _____ Student #: _____ Student Email #: _____

1. Quotient Rule

Compute: $\frac{d}{dx} \left(\frac{\cos x}{\sin x} \right)$

2. Chain Rule

Compute: $\frac{d}{dx} \ln(\cos(x))$

3. Product Rule + Chain Rule

Compute: $\frac{d}{dx} (\cos(x) \sin(x) - \frac{1}{2} \sin(2x))$

4. Chain Rule + Product Rule

Compute: $\frac{d}{dx} (e^{2x \cos(x)})$